

PURNIMA BANK

SentinelAI Fraud Detection Platform

BUSINESS REQUIREMENTS DOCUMENT

SentinelAI Fraud Detection Platform

Business Requirements – Real-Time Card Fraud Detection

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Related Document:	Project Charter v1.0

1. Introduction

This Business Requirements Document (BRD) captures the business needs driving the SentinelAI Fraud Detection Platform and translates the objectives in the Project Charter into requirements that the solution must satisfy. It is the baseline for the Functional Requirements Document (FRD) and for UAT test case design.

2. Business Background

Fraud Operations currently reviews approximately 9,000 flagged transactions per day, of which fewer than 15% are confirmed fraud. Analysts report escalating alert fatigue, and the false-decline rate is now a top-5 driver of card attrition per the Q4 2025 Customer Experience survey. The business requires a fraud detection capability that materially improves both loss prevention and the customer experience simultaneously.

3. Business Requirements

3.1 Fraud Detection & Scoring

ID	Requirement	Priority	Source
BR-01	The system shall generate a real-time fraud risk score (0–999) for every card-present and card-not-present transaction prior to authorization.	Must	Fraud Ops
BR-02	The system shall return a fraud score within 200ms at the 95th percentile to avoid impacting authorization SLAs.	Must	Card Payments
BR-03	The system shall support configurable score thresholds per card product and merchant category code (MCC).	Must	Fraud Ops
BR-04	The system shall incorporate device, geolocation, and behavioral biometrics signals where available.	Should	Data Science
BR-05	The system shall reduce the false-positive decline rate to 4.0% or lower.	Must	CRO

3.2 Case Management & Analyst Workflow

ID	Requirement	Priority	Source
BR-06	The system shall auto-generate a case in the existing case management tool for any transaction scoring above the configured threshold.	Must	Fraud Ops
BR-07	The system shall provide top-5 contributing reason codes for every generated case to support analyst review.	Must	Fraud Ops / Compliance
BR-08	The system shall allow analysts to submit case dispositions that feed back into model retraining datasets.	Should	Data Science
BR-09	The system shall reduce manual case review volume by at least 30% within 90 days of go-live.	Should	CRO

3.3 Regulatory & Governance

ID	Requirement	Priority	Source
BR-10	The model shall produce explainable, human-readable reason codes suitable for Reg E dispute correspondence.	Must	Compliance
BR-11	The model development lifecycle shall comply with the Bank's Model Risk Management policy (aligned to SR 11-7).	Must	Compliance
BR-12	The system shall not use protected-class attributes (or close proxies) as model features.	Must	Legal / Compliance
BR-13	The system shall maintain a full audit trail of scores, decisions, and overrides for a minimum of 7 years.	Must	Compliance

3.4 Reporting & Monitoring

ID	Requirement	Priority	Source
BR-14	The system shall provide a real-time executive dashboard of fraud loss, false-positive rate, and case volume.	Must	CRO
BR-15	The system shall alert the Data Science team when model performance (precision/recall) drifts beyond defined thresholds.	Must	Data Science

4. Assumptions & Dependencies

- Requirements assume continued access to 24+ months of labeled historical fraud data.
- Dependent on the Card Payments team's ability to expose a real-time scoring hook in the authorization switch.
- Dependent on Compliance's model risk validation timeline, which is externally resourced.

5. Out of Scope

- Wire and ACH fraud detection
- Merchant chargeback dispute automation
- Customer-facing mobile fraud alerts (Phase 2)

6. Approval

Name / Role	Approval	Date
David Okonjo – Head of Fraud Operations	Approved	Feb 3, 2026
Priya Nataraj – Head of Data Science	Approved	Feb 3, 2026
Sarah Whitfield – Chief Risk Officer	Approved	Feb 4, 2026